

Computer assignment 2

ESG and life insurance

Begränsad delning

Introduction

In this assignment the focus of the first half is on building a simulation model of “the market”, here consisting of US zero coupon bond interest rates and ten US stocks. The models considered are a VAR(1), historical simulation (HS), and an AI-generated model.

Use weekly logarithmic return data for calibration and then simulate 5000 scenarios with 480 weekly time steps for each model. These scenarios will be used in the next computer assignment for calculating risk capital and the impact of different investment strategies for a life insurance company.

You can find the calibration data (for the period 1996-01-01 -> 2025-12-31) in the file *market_data.csv* on Canvas where the columns are weekly time series of stock prices and (US government) interest rates in the following order:

1. Boeing (BA)	9. Walmart (WMT)
2. Caterpillar (CAT)	10. Exxon Mobile (XOM)
3. Coca Cola (KO)	11. 5Y rate
4. Walt Disney (DIS)	12. 10Y rate
5. JP Morgan (JPM)	13. 15Y rate
6. 3M (MMM)	14. 20Y rate
7. Microsoft (MSFT)	15. 30Y rate
8. Pfizer (PFE)	

The second half of the assignment is concerned with mortality estimation and calculation of the first order reserve. This will also be used in the next assignment.

Important: All code in this assignment must be written by yourselves, except in 4 where AI-code is encouraged.

1. Explore the weekly data set (market_data.csv)

- Plot the evolution of the stocks and the interest rates over the period 1996-01-01 -> 2025-12-31, in two separate plots (with legends). Normalize all the stock prices so that they have an initial price of 100 enabling a fair comparison of their performances. Do not normalize the rates.
- Determine for each stock (and present in a table):
 - the annualized mean log-return,
 - the annualized volatility of log-returns,
 - the minimum and maximum log-return.
- Determine the correlation matrix for the log-returns of all stocks and interest rates and plot as a heatmap, together with the values formatted as percent.

2. VAR(1)-model

Estimate a VAR(1)-model for the market.

- Estimate the parameters (and include them in the report)
- Show that the model is stable.
- Simulate 5000 scenarios with 480 weekly time steps for all assets and plot the price (or interest rate) percentiles (5%, 10%, 25%, 50%, 75%, 90%, 95%) for the stocks Caterpillar and Walmart and the 5- and 30-year interest rates. A total of four plots. Also list these percentiles, for the time periods 240 and 480, in two tables with the percentiles in the horizontal direction and the stocks and interest rates in the vertical direction.

Discussion: Do you observe any unrealistic behavior in the plots?

3. Historical simulation

Use basic historical simulation to simulate 5000 scenarios with 480 weekly time steps for all assets and plot the price (or interest rate) percentiles (5%, 10%, 25%, 50%, 75%, 90%, 95%) for the stocks Caterpillar and Walmart and the 5- and 30-year interest rates. A total of four plots. Also list these percentiles, for the time periods 240 and 480, in two tables with the percentiles in the horizontal direction and the stocks and interest rates in the vertical direction. Plot the percentiles (as dashed lines) in the same plot as the VAR(1)-percentiles.

4. AI-generated ESG

Use AI to construct and calibrate a more realistic ESG, preferably based on a time series model or a stochastic differential equation (SDE) model. For a time series model, use a dynamic conditional covariance structure and for an SDE model, include stochastic volatility. Do not use a one factor interest rate model.

- In the report, include which AI-tool you used, the prompt and a short description of your model and the calibration
- Do 2.c) again and plot the price (or interest rate) percentiles (as dotted lines) in the same plot as the VAR(1)- and HS-percentiles

Discussion: How does your AI ESG compare to the VAR(1) model and HS? Do the simulated stocks have mean returns, volatilities and correlations like the ones in the data?

LIFE INSURANCE

We assume that the company has a single policyholder, a 60-year-old female (born 1965-12-24) who will retire at 65 with lifelong payments and no survivor benefits. This policy has a guaranteed interest rate of 2% and at the calculation date (2025-12-31) we have:

Status	PP
Premium	120 000
G(0)	2 000 000

Collect data for mean populations and number of deaths for men and women (separately) from statistikdatabasen at SCB: [Statistikdatabasen - Välj tabell \(scb.se\)](https://statistikdatabasen.scb.se) for 2021 – 2025.

Befolkning->Befolkningsstatistik->Medelfolkmängd->Medelfolkmängd (efter födelseår) efter region, ...

Befolkning->Befolkningsstatistik->Döda->Döda efter region, ålder (efter födelseår) och kön

5. Explore the data

- a) Plot the average, over the five years of data, of the mean population for each age for both men and women in the same figure.
- b) Plot the average, over the five years of data, of the number of deaths for each age for both men and women in the same figure.

Discussion: What do these plots tell you?

6. Mortality estimation

To find the Makeham parameters, use the method from the lecture.

- a) Estimate the force of mortality for men and women separately using the SCB-data and plot these estimates in the same figure. Do this for ages 20 – 99 and use data for all five years for a more stable estimate. Include a table of these estimates for ages 60, 70, 80 and 90.
- b) Fit Makeham functions¹ using only force of mortality estimates for ages 20 – 90 from above. Then plot the estimated forces of mortality, for ages 20 – 99, together with the Makeham functions in the same figure (both men and women). Include the parameters in the report.
- c) Plot the survival functions for men and women, for ages 0 – 105, in the same figure (but not the same as in b).
- d) Calculate life expectancy for both men and women (both at birth and at age 65), according to the lifetime distributions given by the Makeham functions.

7. 1st order payout and reserve

Calculate the guaranteed payout amount and project the 1st order reserve for the policy.

- a) Determine the yearly payout amount, b .
- b) Plot the evolution of the reserve 30 years into the future.
- c) Include a table with the value of the reserve for every 5 years.

¹ Using the method from the lecture

Grading

These are the requirements for the grades 3, 4 and 5:

Problem	3	4	5
1	All	All	All
2	All	All	All
3	-	All	All
4	-	-	All
5	All	All	All
6	All	All	All
7	All	All	All